

SQL Learning Roadmap (Beginner → Interview Ready)

Phase 0 – Database Fundamentals (Don't Skip)

Learn these concepts before writing SQL:

- What is a Database?
 - What is an RDBMS?
 - Tables
 - Rows & Columns
 - Schema
 - Data Types
 - NULL values
 - Primary Key
 - Foreign Key
 - Constraints
 - NOT NULL
 - UNIQUE
 - PRIMARY KEY
 - FOREIGN KEY
 - DEFAULT
 - CHECK
-

Phase 1 – Basic SQL Queries

Master these first:

- SQL Syntax
 - SELECT
 - DISTINCT
 - Aliases (AS)
 - LIMIT / TOP
 - ORDER BY
-

Phase 2 – Filtering Data

Learn:

- WHERE

- Comparison Operators
 - Logical Operators (AND, OR, NOT)
 - BETWEEN
 - IN
 - LIKE
 - IS NULL
 - IS NOT NULL
-

Phase 3 – Aggregate Functions

Learn:

- COUNT()
 - SUM()
 - AVG()
 - MIN()
 - MAX()
-

Phase 4 – Grouping Data

Learn:

- GROUP BY
- HAVING

This is where analytical SQL begins.

Phase 5 – SQL Joins (Very Important)

Master every join.

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL OUTER JOIN
- CROSS JOIN
- SELF JOIN

Also revise:

- Primary Keys
- Foreign Keys

Phase 6 – Conditional Logic

Learn:

- CASE WHEN
- COALESCE()
- NULLIF()

Phase 7 – Data Manipulation (DML)

Learn:

- INSERT
- UPDATE
- DELETE

Understand how data is modified.

Phase 8 – Database Objects (DDL)

Learn:

- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- TRUNCATE
- RENAME

Phase 9 – Transactions

Learn:

- ACID Properties
 - BEGIN
 - COMMIT
 - ROLLBACK
 - SAVEPOINT
-

Phase 10 – Indexes

Understand:

- What indexes are
 - Why indexes improve performance
 - Clustered vs Non-Clustered (basic understanding)
-

Intermediate SQL

Phase 11 – Subqueries

Learn:

- Subqueries in WHERE
 - Subqueries in SELECT
 - Subqueries in FROM
 - Correlated Subqueries
 - EXISTS
 - NOT EXISTS
 - ANY
 - ALL
-

Phase 12 – Common Table Expressions (CTEs)

Learn:

- WITH clause
 - Multiple CTEs
 - Recursive CTE (basic understanding)
-

Phase 13 – Window Functions (Must Know)

Master:

- OVER()
- PARTITION BY
- ORDER BY inside OVER()
- ROW_NUMBER()

- RANK()
 - DENSE_RANK()
 - NTILE()
 - LAG()
 - LEAD()
 - FIRST_VALUE()
 - LAST_VALUE()
 - Running Totals
 - Moving Average
-

Phase 14 – Advanced SQL Concepts

Learn:

- UNION
 - UNION ALL
 - INTERSECT
 - EXCEPT / MINUS
 - Views
 - Temporary Tables
 - Common SQL Functions
 - String Functions
 - Date Functions
 - Numeric Functions
 - Conversion Functions
-

Phase 15 – Query Optimization (Interview Basics)

Learn:

- Execution Plan (basic)
 - Index Usage
 - When NOT to use SELECT *
 - Join Optimization
 - Filtering Early
 - Common Performance Tips
-

SQL Interview Topics

Before interviews, revise these thoroughly:

- Joins
 - GROUP BY vs Window Functions
 - HAVING vs WHERE
 - DELETE vs TRUNCATE vs DROP
 - UNION vs UNION ALL
 - RANK vs DENSE_RANK vs ROW_NUMBER
 - Primary Key vs Unique Key
 - Primary Key vs Foreign Key
 - Clustered vs Non-Clustered Index
 - EXISTS vs IN
 - CTE vs Subquery
 - Window Functions
 - Transactions & ACID
 - Normalization (1NF, 2NF, 3NF)
 - Common SQL Functions
-

Learning Order

1. Database Fundamentals
2. Basic Queries (SELECT)
3. Filtering (WHERE)
4. Aggregate Functions
5. GROUP BY & HAVING
6. Joins
7. CASE & NULL Handling
8. INSERT / UPDATE / DELETE
9. CREATE / ALTER / DROP
10. Transactions
11. Indexes
12. Subqueries
13. CTEs
14. Window Functions
15. UNION & Views
16. SQL Functions (String, Date, Numeric)
17. Query Optimization
18. Interview Questions & Practice